ADITYA UNIVERSITY

(Formerly known as Aditya Engineering College (A))

000407

PART-I

S.No: 421947

Signature of the Controller of Examination	J. Pavan.
Signature of the Student with date	U. Umamahesh 18/11/25
Signature of the Invigilator with date	M. Sai Chaitanya 18/11/25
Serial No. Last Page Written	3

Programe : B.TECH

Hall Ticket No: 22A91A0348

Name : UNGARALA UMA MAHESWARARAO

Examination : III SEMESTER SUPPLEMENTARY (AR20(R))

Month-Year : Nov/Dec 2025

Branch : MECHANICAL ENGINEERING

Course Code : 201BS3T12

Course Name : Integral Transforms And Applications Of PartialDifferential Equations

Date of Exam : 18-11-2025

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   - i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   - ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   - iii. Writing religious symbols.
   - iv. Either seeking or providing any assistance to the fellow students in the examinations.
   - v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   - vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
   - vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

Q.No.

Unit - I

1) a) Given data

The Fourier Series Of $f(x)$

$$= \begin{cases} x & \text{if } 0 \leq x \leq 3 \\ 6-x & \text{if } 3 \leq x \leq 6. \end{cases}$$

Fourier Series of the $f(x)$

Q.No.

Q.No.

Q.No.

Q.No.

Q.No.

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~~Aditya Engineering College.~~

~~Aditya University~~

~~Aditya Autonomous~~

~~Aditya Engineering College And Technology~~

~~Sir Raman Bhavan. to Ramanujan Bhavan~~

~~Using Fourier Integral Show that Find The Half - Range Fourier Cosine Series Of $f(x) = x$ In The Interval Dt by Using Laplace Transform Find Using Convolution Therom Find The Finit Fourier Sine and Cosine Tranforms The Half Rang Solve the Equation With Bondary Condition Where Solve Ginen By Using The Method of Separation Of Variables By Using The Laplace Transform Solve.~~

SEMI-LOG PAPER (5 CYCLES X 1/10")